

Assignment 3

1. Prove that the quotient group $\mathbb{R}/\mathbb{Z}$ is isomorphic to the circle (with respect to multiplication).
2. Let H and K be subgroups of a group G of finite index. Show that $H \cap K$ also has finite index.
3. Let G be a finite group and H and K are subgroups of G . Prove that :

$$|HaK| = \frac{|H||K|}{|H \cap aKa^{-1}|}$$

for all $a \in G$.

4. Suppose H is a subgroup of G such that whenever $Ha \neq Hb$, then $aH \neq bH$. Show that H is normal in G .
5. Let G be a finite abelian group such that the number of solutions of $x^n = e$ is at most n for every positive integer n . Show that G is cyclic.